



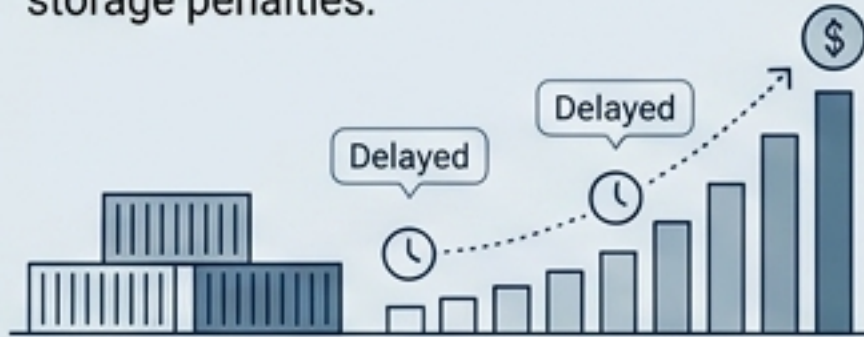
FreightX

Autonomous SME Trade Finance & Cargo
Passports on the Arc Network.

Financial

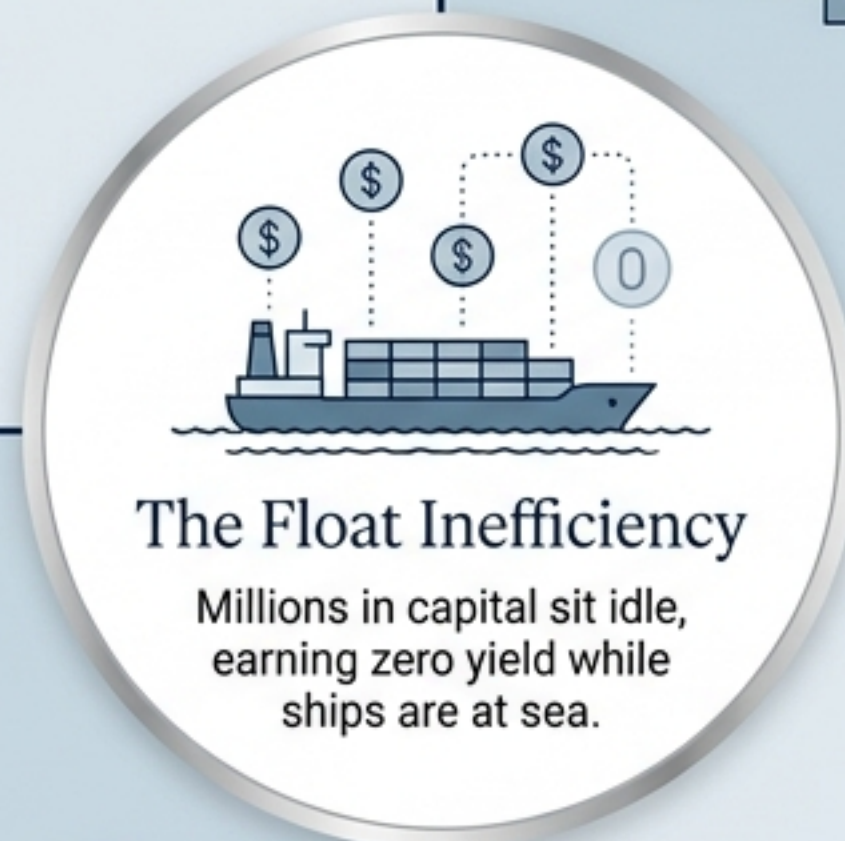
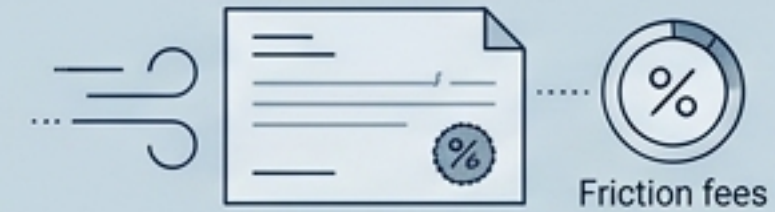
Demurrage Extortion

Manual cargo retrieval tracking triggers astronomical, delayed, and easily manipulated port storage penalties.



Working Capital Deadlock

SME suppliers lock \$30K–\$500K for 30–90 days. Sluggish bank Letters of Credit extract 1.5%–3% in pure friction fees.

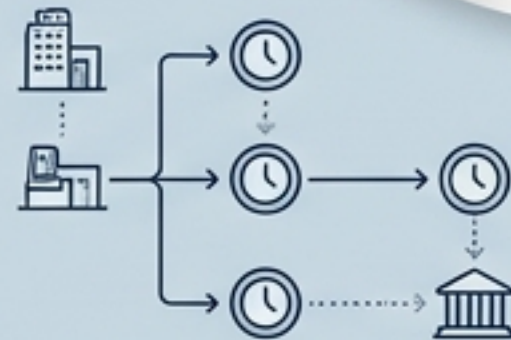


Manual

Opaque

Paper Bill-of-Lading Workflows

Reconciliation across shippers, ports, and customs takes weeks using physical, forgeable paper slips.

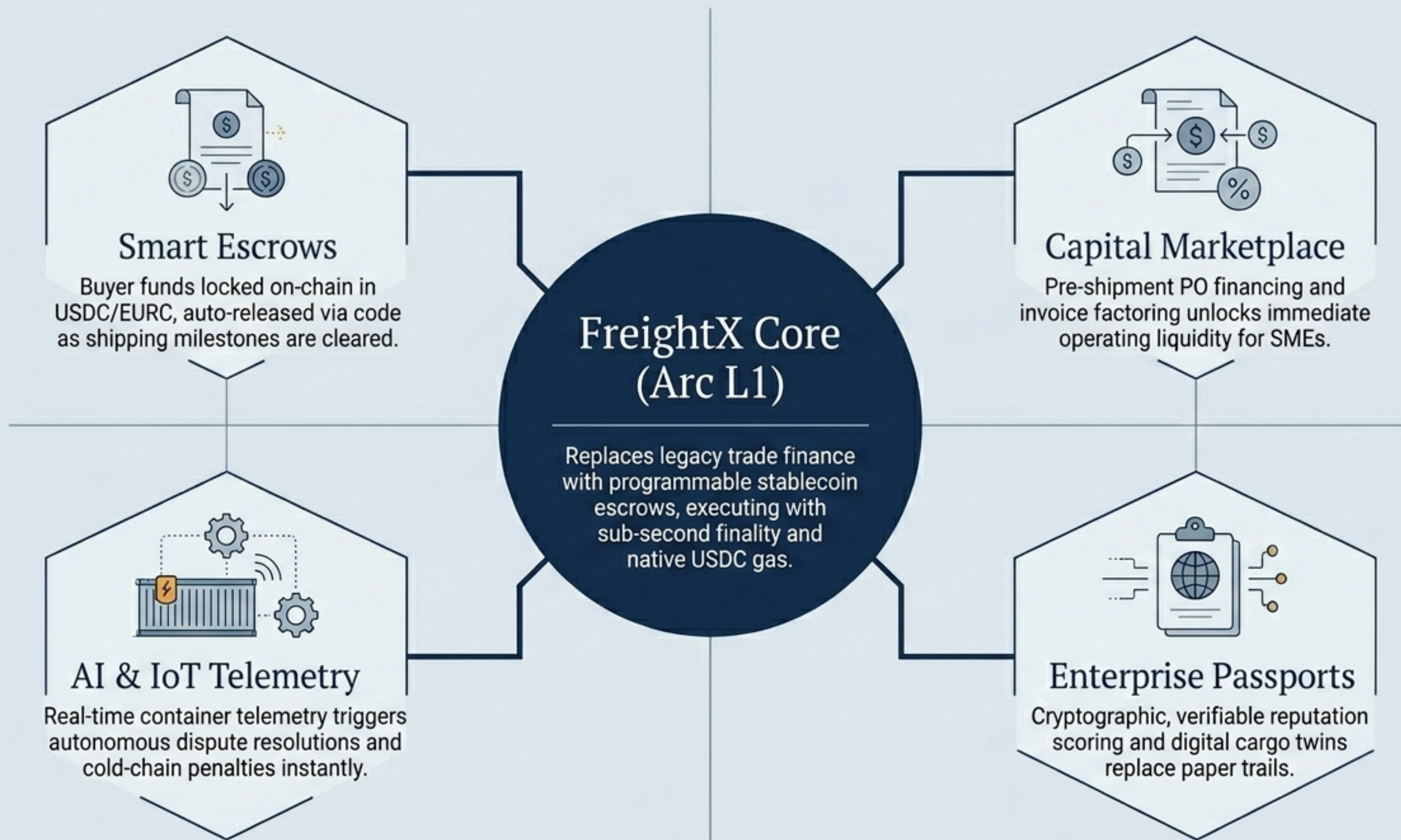


Cold-Chain Vulnerability

Unverified temperature conditions ruin high-value perishables, leading to months of locked-capital legal disputes.



Operational



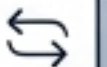
Dimension	Legacy Trade System	FreightX Arc-Powered Solution
Asset Tracking	Paper Bill of Lading	ERC-721 Digital Twin Passport
Trade Finance	Bank Letters of Credit (5-15 days)	Programmable Smart Escrow (Instant)
Working Capital	Locked & Idle for weeks	USYC Swept (5% APY active yield mid-transit)
Dispute Resolution	Manual Arbitration	Automated IoT Telemetry & Immutable Penalties
Currency & Settlement	Fragmented, volatile FX	USDC / EURC on Arc L1 with sub-second finality

Circle Stack Integrations

USDC / EURC 
Native gas & multi-currency settlement.

USYC 
Tokenized treasury for yield sweeping.

Developer Wallets 
Powering AI Agent oracles. 

StableFX 
Integrated AED-to-USD routing.

User Interface
Next.js 15 App Router & React 19
Frontend with Server-Sent
Events for real-time telemetry.

Web3 Middleware
Viem v2.50 & Wagmi v2.19 providing EVM
RPC transport. Next.js Edge routing.

Arc L1 Protocol
Custom Solidity 0.8.20 smart contracts
deployed on Arc Testnet.

Phase 1: Pre-Shipment PO Financing & Invoice Factoring

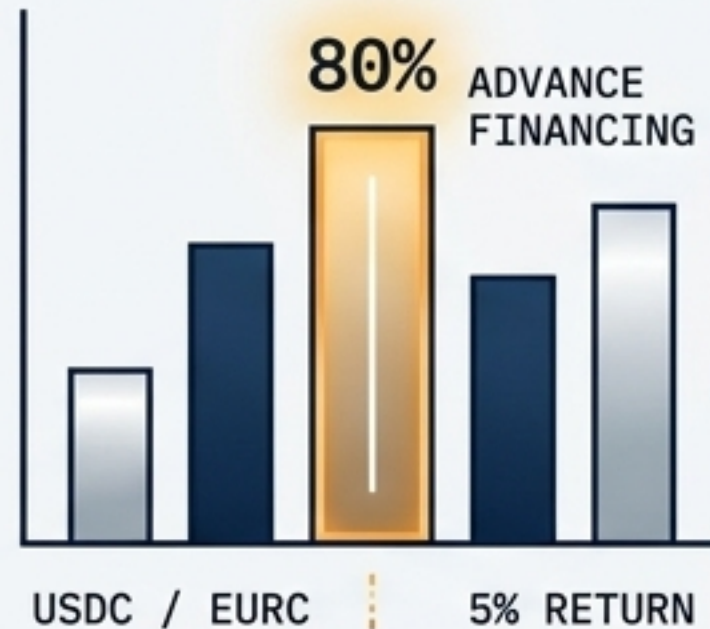
Step 1: The Request

Suppliers list Purchase Orders or pending cargo invoices on the Marketplace to unlock production liquidity.



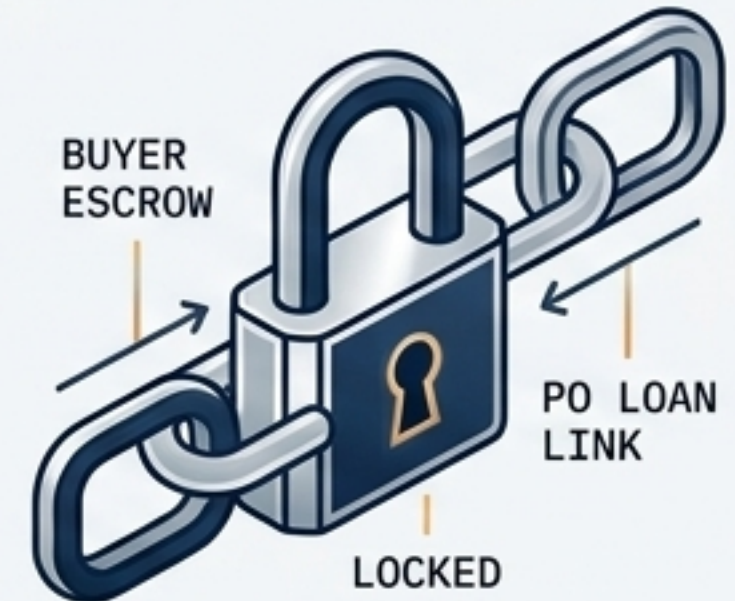
Step 2: The Funding

External investors fund POs instantly in USDC/EURC, securing up to 80% advance financing for a fixed 5% return.



Step 3: Automated Repayment

Buyer escrow deposits automatically link to the active PO loan, locking investor repayment into the final settlement.



Phase 2: Cargo Digital Twins & Smart Escrows

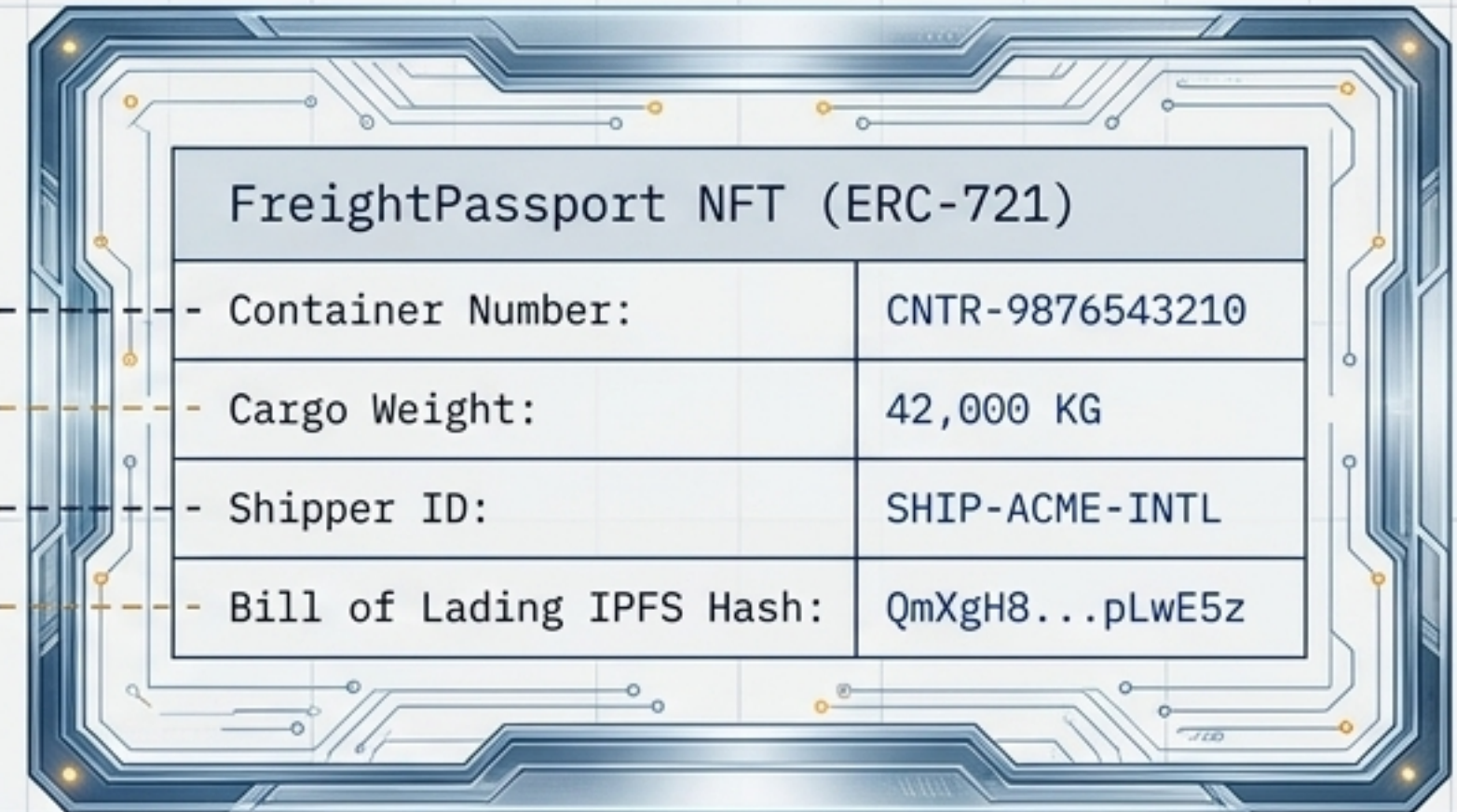
The Physical



The Escrow Lock

Buyers book the shipment and lock total cargo value plus shipping fees into the **FreightEscrow** contract.

The Digital



The Digital Twin Mint

An **ERC-721 FreightPassport NFT** is minted concurrently, acting as the immutable digital twin.

The Capital Engine: Sweeping Idle Treasuries

The Float Inefficiency

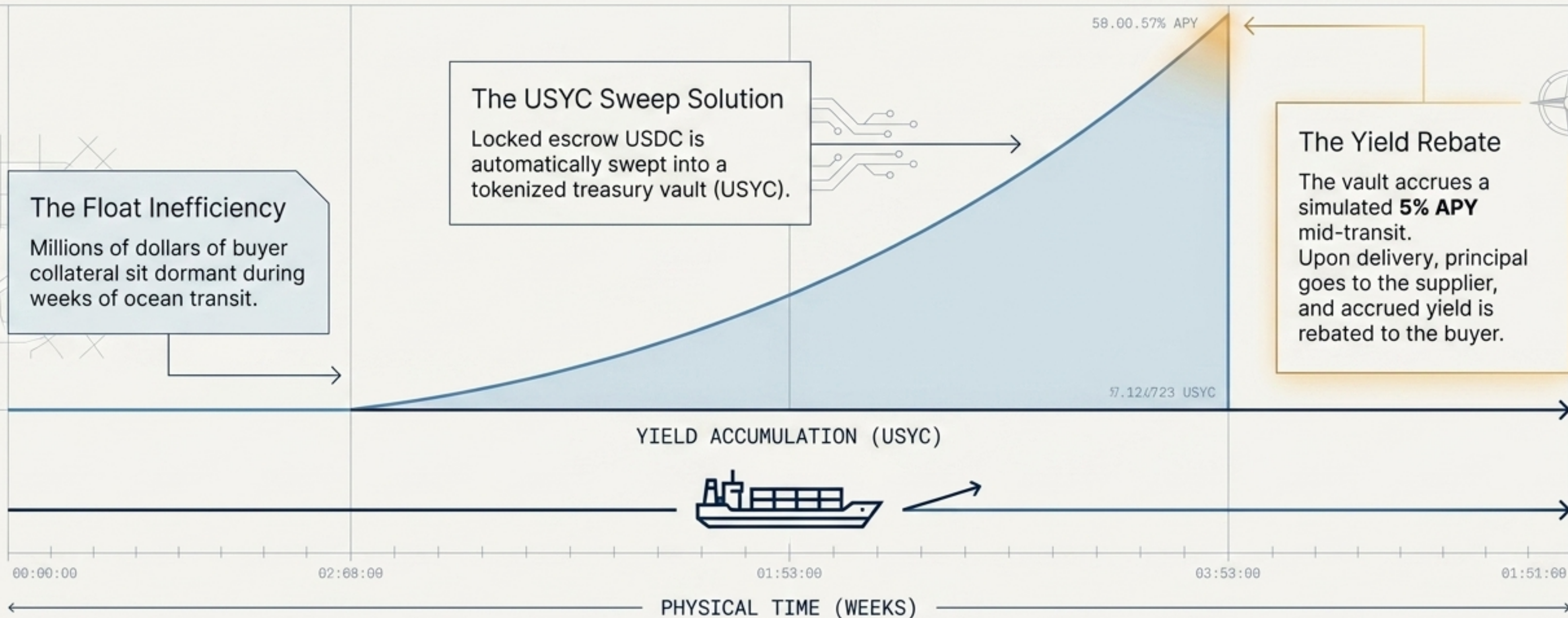
Millions of dollars of buyer collateral sit dormant during weeks of ocean transit.

The USYC Sweep Solution

Locked escrow USDC is automatically swept into a tokenized treasury vault (USYC).

The Yield Rebate

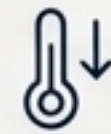
The vault accrues a simulated **5% APY** mid-transit. Upon delivery, principal goes to the supplier, and accrued yield is rebated to the buyer.

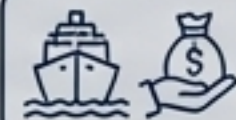


Phase 3: Autonomous IoT Telemetry & Nanopayments

Telemetry Feed

TIMESTAMP	SENSOR_ID	GPS_COORDS	TEMP (°C)	HUMIDITY (%)	STATUS
2024-10-26 14:30:15	IOT-SENS-001	01°17'N, 103°51'E	5.8	45	OK
2024-10-26 14:35:20	IOT-SENS-002	01°17'N, 103°51'E	6.1	46	OK
2024-10-26 14:40:25	IOT-SENS-001	01°17'N, 103°52'E	6.9	48	OK
2024-10-26 14:45:30	IOT-SENS-003	01°17'N, 103°52'E	8.2	50	TEMP_BREACH

 **Thermal Breach Penalties:** Sensors detect a cold-chain breach (>8.0°C). Smart contract automatically applies a predefined 5% deduction to the supplier's payout.



Mid-Transit Milestones: GPS confirms arrival at the Singapore Transshipment Hub. Contract instantly releases 30% of locked funds to the supplier.

AI Logic & Transactions



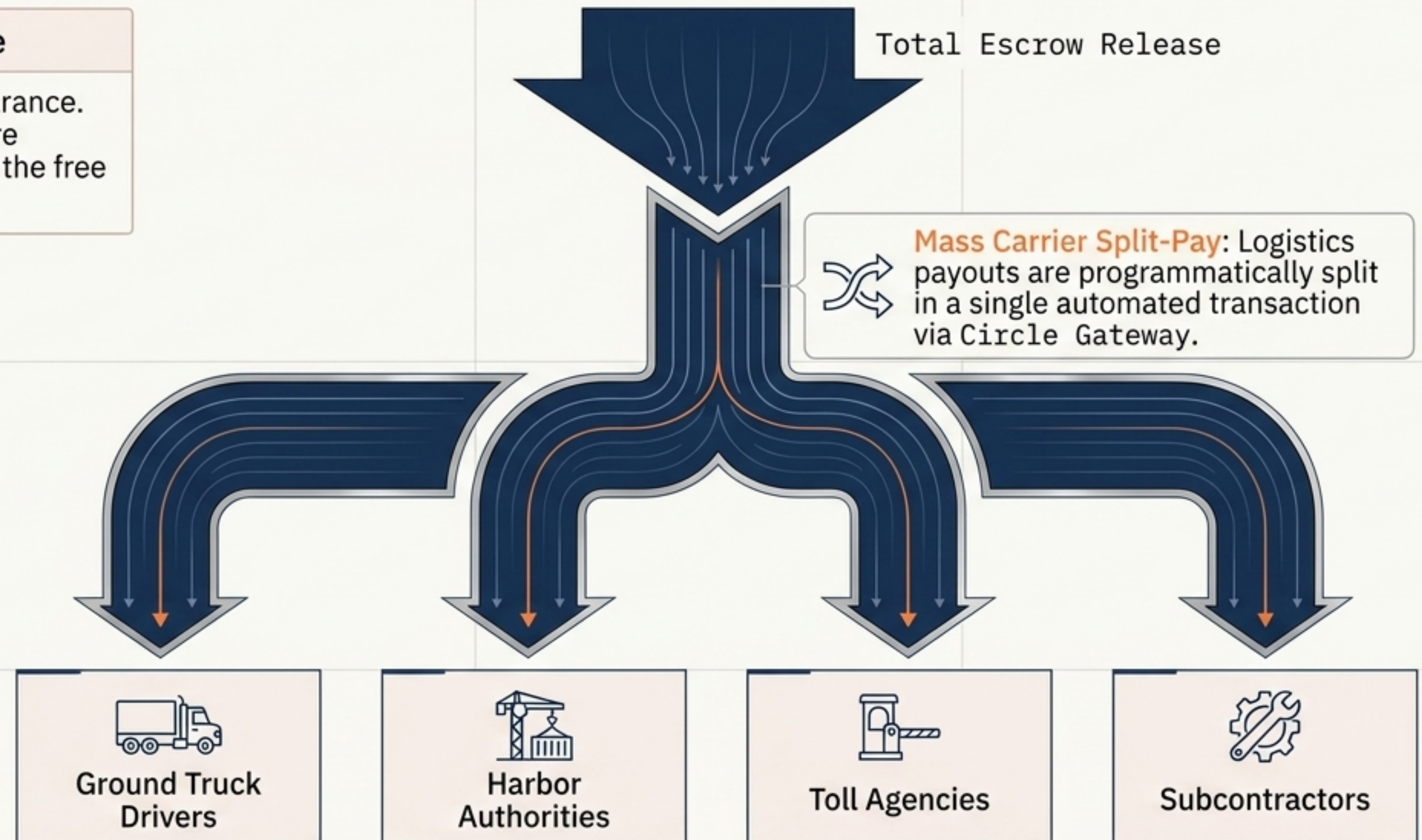
AI Logistics Coordinator: An autonomous agent executes sub-cent USDC nanopayments to unlock server-side IoT telemetry logs (HTTP 402 Paywalls).

```
1 // AI Logistics Coordinator: Autonomous Nanopayment Execution
2 POST /api/v1/telemetry/logs/access
3 Headers: {
4   "Authorization": "Bearer [AI_AGENT_TOKEN]",
5   "Content-Type": "application/json"
6 }
7 Body: {
8   "containerId": "CNTR-9876543210",
9   "logType": "FULL_TELEMETRY"
10 }
11
12 <-- RESPONSE 402 Payment Required
13 {
14   "error": "HTTP 402 Payment Required",
15   "message": "Access to full logs requires micropayment.",
16   "amount": 0.005,
17   "currency": "USDC",
18   "paymentAddress": "0x[SMART_CONTRACT_PAYWALL_ADDRESS]"
19 }
20
21 // AI Agent triggers automated payment
22 TX_HASH = sendNanopayment("0.005", "USDC", "0x[SMART_CONTRACT_PAYWALL_ADDRESS]");
23
24 --> SUCCESS 200 OK (Access Granted)
25 {
26   "status": "SUCCESS",
27   "message": "Payment verified. Access granted.",
28   "logData": { ... }
29 }
```

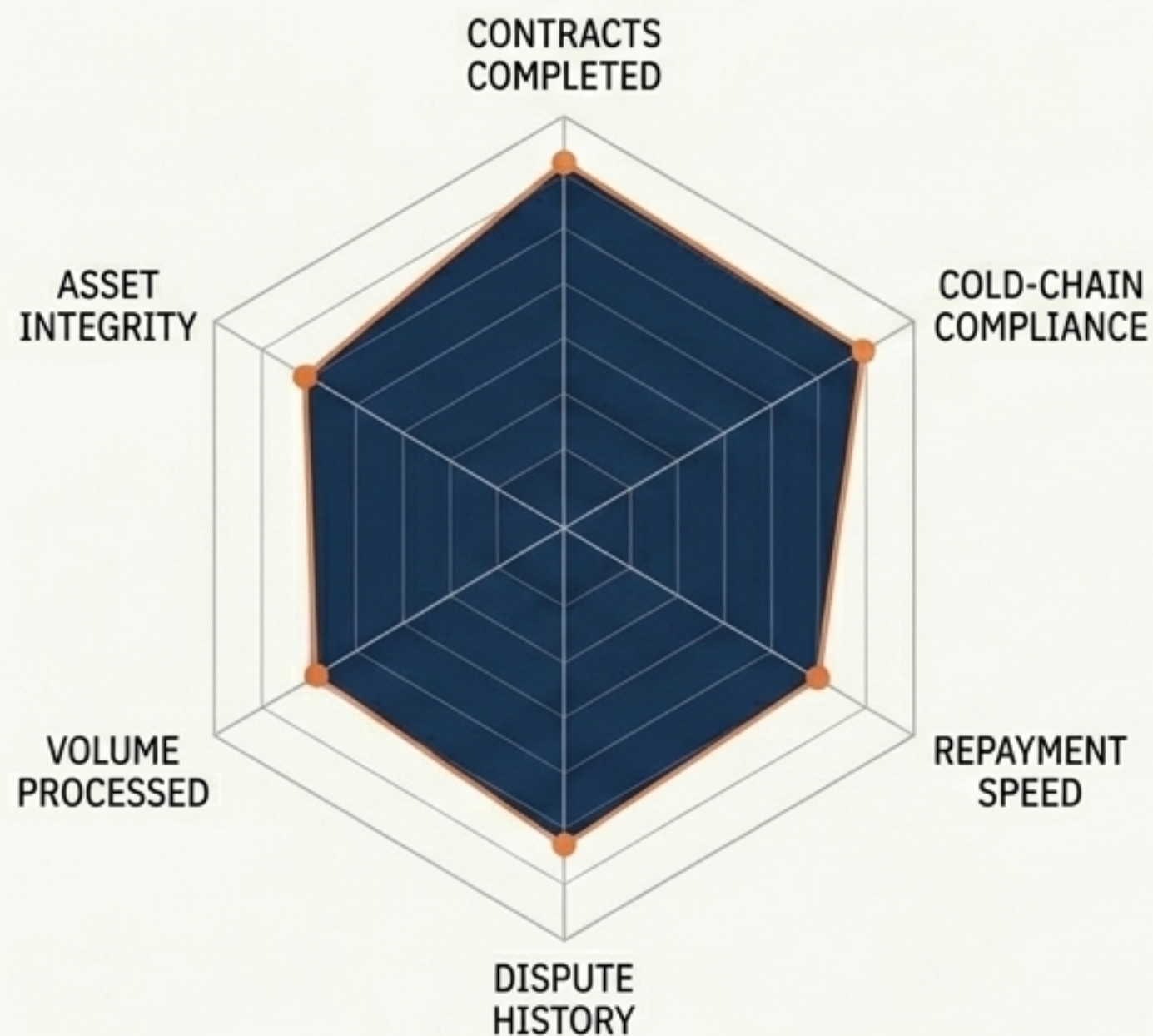

Phase 4: Demurrage Acceleration & Settlement Waterfall

Time-Accelerated Demurrage

Timers start upon Customs Clearance. Progressive storage penalties are auto-deducted if cargo exceeds the free discharge window.



Cryptographic Reputation Passports



On-Chain Analysis

Protocol analyzes metadata (completed contracts, cold-chain compliance, repayment speed) to generate dynamic enterprise grades.

```
{
  "@context": [
    "https://www.w3.org/2018/credentials/v1",
    "https://www.w3.org/2018/credentials/examples/v1"
  ],
  "id": "http://example.edu/credentials/3732",
  "type": ["VerifiableCredential", "ReputationPassport"],
  "issuer": "did:example:logisticsDAO:1234",
  "issuanceDate": "2024-10-27T19:23:24Z",
  "credentialSubject": {
    "id": "did:example:carrier:5678",
    "reputationScore": "94.5",
    "coldChainCompliance": "98%",
    "completedContracts": "150+"
  },
  "proof": {
    "type": "RsaSignature2018",
    "created": "2024-10-27T19:23:24Z",
    "proofPurpose": "assertionMethod",
    "verificationMethod": "did:example:logisticsDAO:1234#key-1",
    "jws": "eyJhbGciOiJIUzUuMiIsInR5cCI6IkpXLTJRCiIsImZsc20sI0lsijY8I119..."
  }
}
```

W3C Verifiable Credentials

1-click export of reputation as a W3C-compliant Verifiable Credential (JSON-LD format) to prove trade history off-chain.

Developer Experience: Building on Arc L1



What Worked Exceptionally Well

- **Deterministic Gas**
Arc's dollar-denominated USDC gas eliminated volatility, making transaction costs highly predictable.
- **Sub-Second Finality**
Speeds perfectly matched the immediate pacing required for physical freight processing.



Circle Product Feedback & Requests

- **Unified Faucets**
A single testnet faucet dispensing all Circle assets (USDC & EURC) would accelerate multi-currency testing.
- **StableFX Sandbox API**
Public sandbox access with mock rates would enhance real-time FX-aware escrow development.

Open Source Governance & Security Safeguards



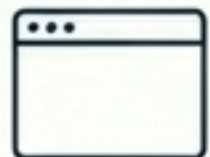
Community Standards

Governed by a strict Contributor Covenant Code of Conduct ensuring a diverse, inclusive developer environment.



Private Key Isolation

Deployer keys are strictly excluded from git tracking.



Browser Sandboxing

Interactive sandbox generates and isolates private keys purely within browser-level localStorage.



Multi-Step Oracles

Escrows require cryptographic signatures from registered IoT gateways before capital disbursements.

The Future Horizon

Waypoint 4: Multi-Sig Arbitration



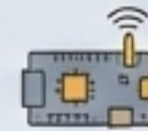
On-chain panel of independent cargo surveyors for complex physical disputes.

Waypoint 3: AI Underwriting



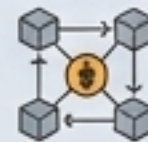
Decentralized credit scoring for optimized factoring discount premiums.

Waypoint 2: Production Telematics



Integrating physical Raspberry Pi Zero/ESP32 cellular sensor nodes.

Waypoint 1: Cross-Chain CCTP



Seamless USDC payment routing across Ethereum, Arbitrum, Solana, and Arc.

Built for the Agora Hackathon — USDC Commerce Stack Challenge

Developed with insight and energy by
Jessica Miller & Team.

GitHub: @jessicamillervi

License: Released under the MIT License.

Explore the live FreightX sandbox on the Arc Testnet.